

AI - Driven Coal Ash Content Estimation Using Photoacoustic Dataset and Synthetic Data Generation

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Project Outline

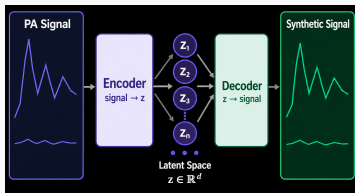
Coal Ash Content Prediction

- ▶ PA Data Acquisition
- ▶ Signal Processing
- ▶ Feature Extraction
- ▶ Feature Ranking
- ▶ Regression
- ▶ ML Model Serialization
- ▶ Development of GUI Deployable Pipeline



Synthetic Data Generation

- ▶ Signal Reconstruction via VAE
- ▶ Latent Feature Extraction
- ▶ Feature Ranking
- ▶ Feature Fusion
- ▶ Development of Decoder
- ▶ Synthetic Data Generation
- ▶ Test Existing Models



Coal Ash → Spontaneous Combustion

1. What is Ash

- ▶ Inorganic, non-combustible mineral residue after coal combustion
- ▶ Comprises oxides: SiO_2 , Al_2O_3 , Fe_2O_3 , CaO , etc.
- ▶ Arises from inherent + extraneous mineral matter

2. Why It Matters

- ▶ Inversely proportional to calorific value
- ▶ Governs coal grading & pricing
- ▶ Causes boiler fouling, slagging, erosion
- ▶ Alters porosity, conductivity & reactivity of coal

3. Spontaneous Combustion Risk

- ▶ Higher ash → lower thermal conductivity → heat retention → hot-spot formation
- ▶ Pyrite (FeS_2) in ash catalyzes & fuels exothermic oxidation
- ▶ Alters pore structure → changes O_2 diffusion into coal matrix
- ▶ Hygroscopic minerals drive moisture-linked heating cycles
- ▶ Ash layer insulates sub-surface coal → traps oxidation heat within stockpile core



State-of-the-Art Methods for Measuring Coal Ash Content

Method	Key Principle
Proximate Analysis	Measures ash residue after complete combustion of coal.
X-ray Fluorescence (XRF)	Determines ash-forming elements using characteristic X-ray emissions.
Prompt Gamma Neutron Activation Analysis (PGNAA)	Uses neutron-induced gamma rays for real-time elemental analysis.
Near-Infrared (NIR) Spectroscopy	Predicts ash content from reflected near-infrared spectra.
Laser-Induced Breakdown Spectroscopy (LIBS)	Uses laser-generated plasma emission for rapid elemental analysis.

Limitations

- ▶ High instrumentation and maintenance cost
- ▶ Frequent calibration requirement
- ▶ Limited suitability for continuous or online monitoring
- ▶ Often relies on complex chemometric models

Research Motivation

- ▶ Existing laboratory techniques generally require **1–3 days** for coal ash content determination, whereas the proposed photoacoustic sensing system can acquire approximately **100 PA signal readings in nearly 2 hours**, enabling rapid data acquisition for AI-assisted ash prediction.

System Overview

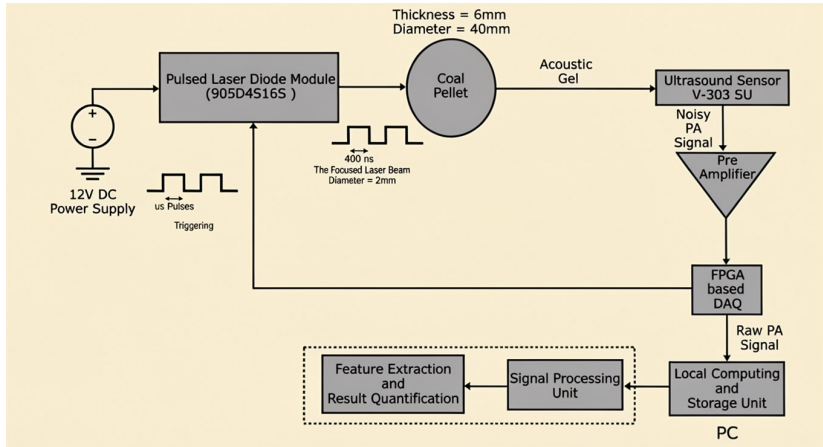
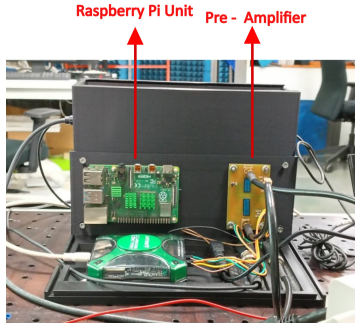
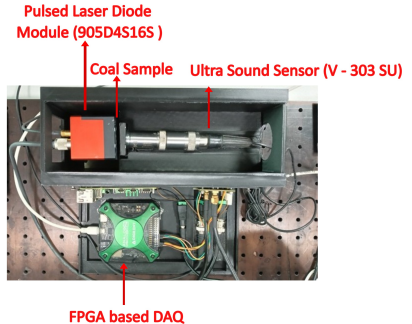


Fig 1 : Block Diagram Representation of the experimental setup of the Photoacoustic Sensing-Based Feature Extractor and ML-Assisted Ash content predictor of coal

Experimental Setup



Side View



Top View

Fig 2 : Top and Side View of the Photoacoustic Sensing based Experimental Setup

Signal Processing

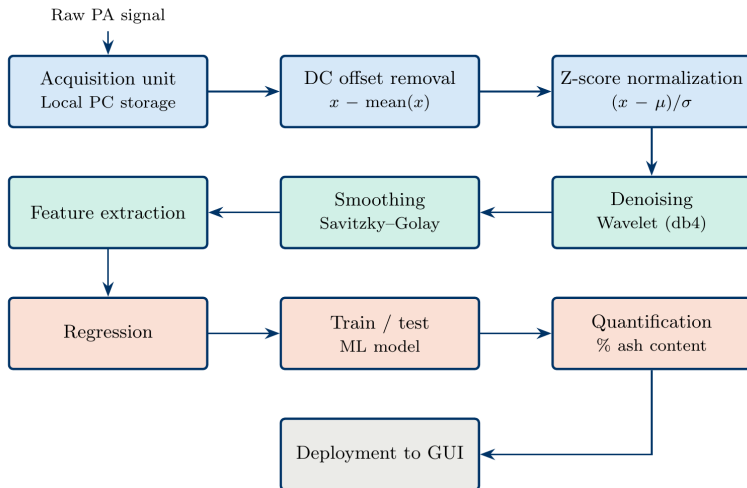


Fig 3 : Block Diagram Representation of Proposed preprocessing, denoising, feature extraction, and ML-assisted Data analysis pipeline for real time Ash content prediction of Coal

Signal Processing

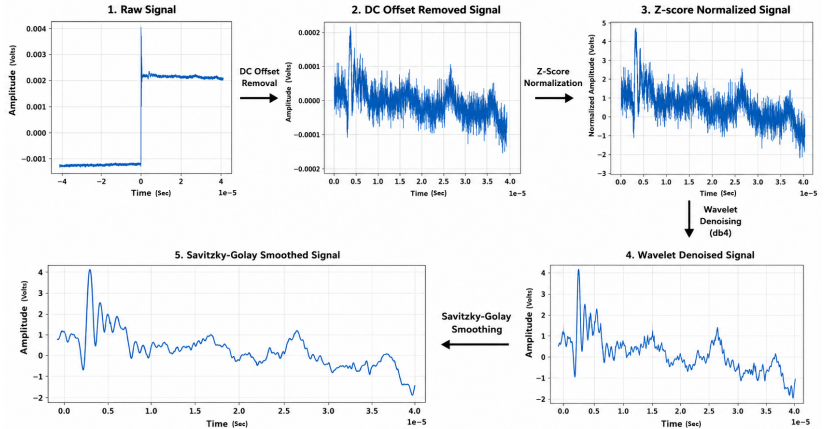


Fig 4 : Signal Processing Stages

Feature Extraction

Processed PA Signal

Time Domain

- ▶ Peak Value
- ▶ Peak Arrival Time
- ▶ Rise Time
- ▶ Fall Time
- ▶ RMS Value
- ▶ Signal Energy
- ▶ Area Under the Curve

Frequency Domain

- ▶ Ratio of the Magnitudes of the 1st and 2nd FFT Peaks

8 Features

Raw PA Signal

Frequency Domain

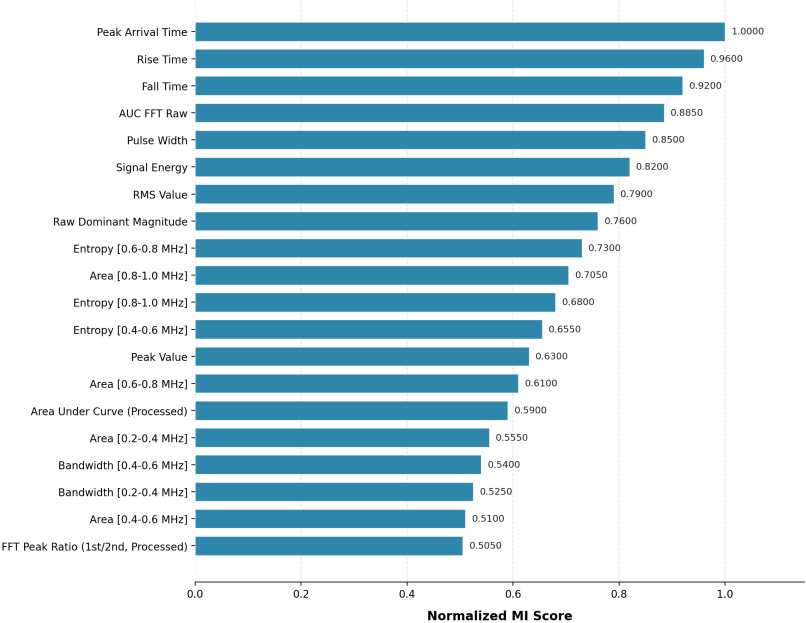
- ▶ Area Under the FFT Curve
- ▶ Magnitude of the Dominant Frequency Component
- ▶ 0.2–0.4 MHz
Area, Bandwidth
- ▶ 0.4–0.6 MHz
Area, Entropy, Bandwidth
- ▶ 0.6–0.8 MHz
Area, Entropy, Bandwidth
- ▶ 0.8–1.0 MHz
Area, Entropy

12 Features

Feature Extraction — Physical Basis

Feature	Physical Reason
Peak Value	Optical absorption strength → drops with ash %
Peak Arrival Time	Lower thermal diffusivity of minerals → delayed peak
Rise Time	Higher thermal mass → slower thermoelastic expansion
Fall Time	Mineral scattering → faster acoustic decay
Energy & AUC	Total absorbed laser energy → decreases with ash %
Total FFT Area	Total spectral energy $\propto$ optical absorption → drops with ash %
Area, Entropy, Bandwidth (at different frequency bands)	Bulk / coarse grain structure, Mid-scale mineral clusters, Fine grain scattering, Surface-level heterogeneity

Feature Ranking



Regression Model Performance

Sl. No.	Model	Training (Mean CV $\pm$ S.D.)	Testing Accuracy
1	XGB Regressor	0.9086 $\pm$ 0.0300	88.81%
2	Extra Trees Regressor	0.8898 $\pm$ 0.0157	89.41%
3	Gradient Boosting Regressor	0.8935 $\pm$ 0.0226	90.21%
4	CatBoost Regressor	0.8940 $\pm$ 0.0170	90.32%
5	SVR-RBF	0.9051 $\pm$ 0.0036	89.35%

Parameter	Value	Parameter	Value
Total Experimental PA Signal Results	900	Total Number of Unique Coal Samples	13
Training Data (80%)	720	Training Data 2D Array Dimension	720 $\times$ 20
Testing Data (20%)	180	Testing Data 2D Array Dimension	180 $\times$ 20

Regression Model

CatBoost Regressor

- ▶ High prediction accuracy with minimal hyperparameter tuning.
- ▶ Effectively captures complex non-linear relationships while reducing overfitting.

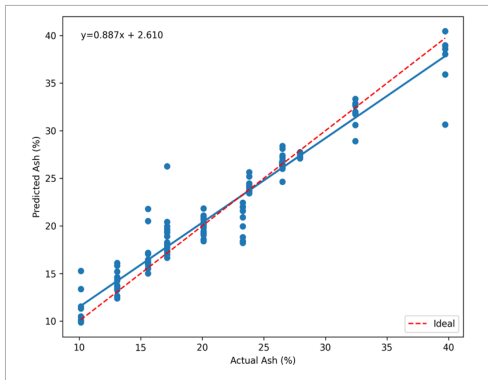
Serialization



model_name.pkl

Extracted Parameters			
Ignition Temperature		Carbon Content	
444.53°C		60.30 %	
Ash Content		Volatile Matter	
23.81 %		15.05 %	

GUI Interface



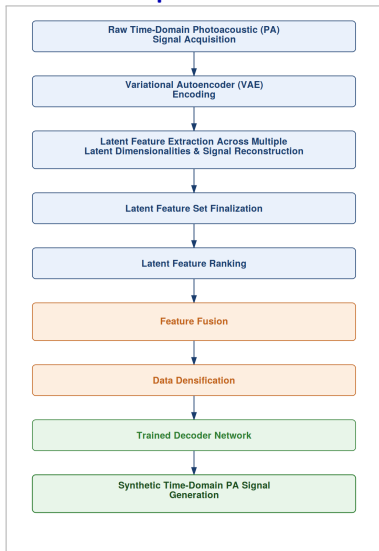
Actual Ash (%) vs Predicted Ash (%)

Regression Equation:

$$y = 0.887x + 2.610$$

Synthetic Data Generation

Synthetic Data Generation Pipeline



Motivation

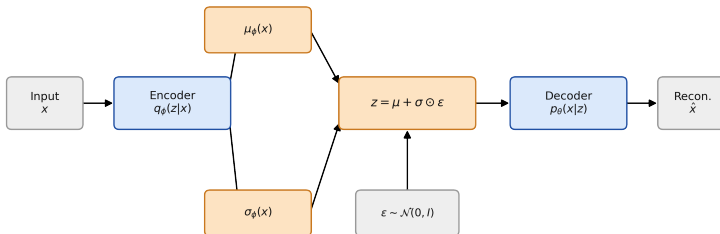
- ▶ Improve model accuracy & generalization
- ▶ Enhance inter-class variability for robust regression
- ▶ Mitigate overfitting under data scarcity

Why VAE

- ▶ Simulation requires accurate physical/acoustic priors
- ▶ Waveform models poorly capture PA source heterogeneity
- ▶ VAE learns data-driven generative latent manifold and captures real sensor noise & setup-specific artifacts
- ▶ Scalable, model-agnostic signal reconstruction

Variational Autoencoder

VAE Architecture



Encoder maps x to a latent distribution; reparameterization enables differentiable sampling; decoder reconstructs x from z .

- ▶ Learns a smooth, continuous, generative latent space
- ▶ Regularizes latent space via KL divergence to enable new-sample generation

Variational Autoencoder

Governing Equations

1. Evidence

Lower

Bound (ELBO)

$$\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)]}_{\text{reconstruction}} - \underbrace{D_{KL}(q_{\phi}(z|x) \parallel p(z))}_{\text{regularization}}$$

2.

Reparameterization
Trick

$$z = \mu_{\phi}(x) + \sigma_{\phi}(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

3. Closed-Form KL
(Gaussian Prior)

$$D_{KL} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

- ▶ Higher β (KL weight) $\rightarrow$ smoother latent space but blurrier reconstructions
- ▶ Higher latent dimension $\rightarrow$ better reconstruction but risk of overfitting/less structured space

Signal Reconstruction → Latent Space Formation

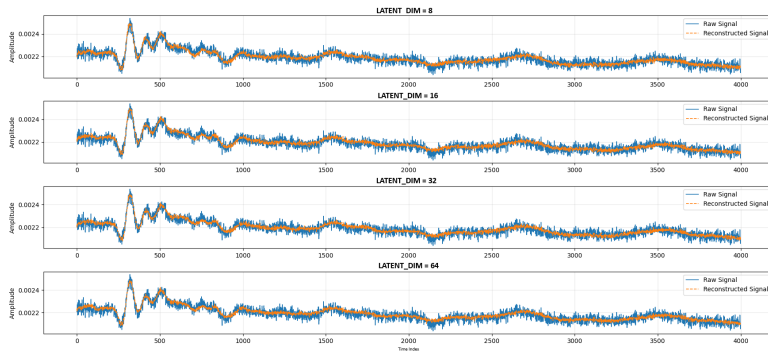


Fig 5 : Signal Reconstruction of Raw Time-Domain Experimental Photoacoustic (PA) Signals Using a Variational Autoencoder (VAE) for Typical Latent Space Dimensions

Latent Feature Ranking

Continuous mutual information between two random variables X and Y

$$I(X; Y) = \iint p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) dx dy$$

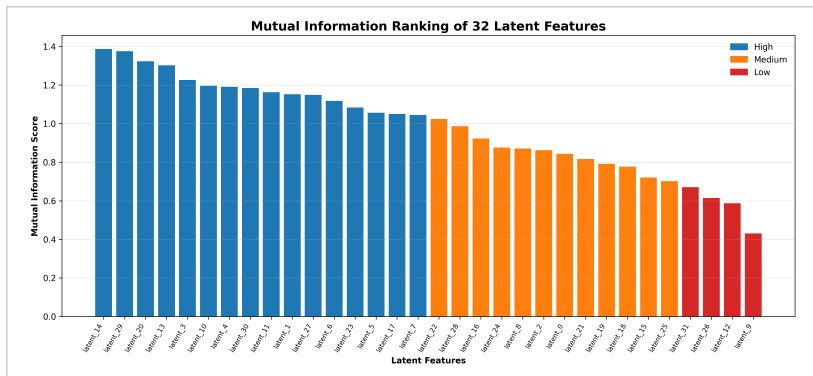
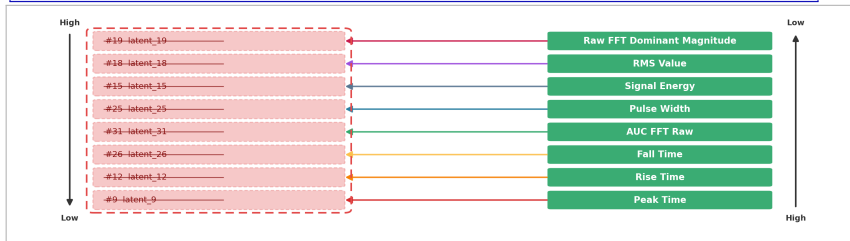


Fig 6 : MI Ranking of 32 Latent Features obtained from Signal Reconstruction

Hybrid Feature Set

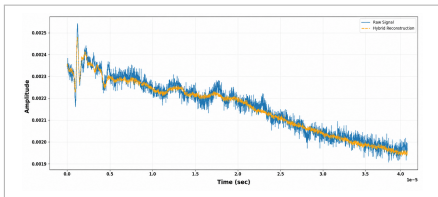
Sl. No.	Latent Features	Engineered Features	Mean Corr.	Mean R^2	Mean Loss (%)
1	16	16	91.22	0.8731	12.69
2	20	12	91.45	0.8773	12.27
3	24	8	91.66	0.8809	11.91
4	28	4	91.71	0.8815	11.85

Hybrid Feature Set of Dimension 32 24 Latent Features + 8 Crafted Features

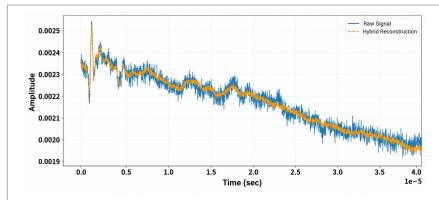


Replacement of the 8 least informative latent features using the 8 most informative engineered features

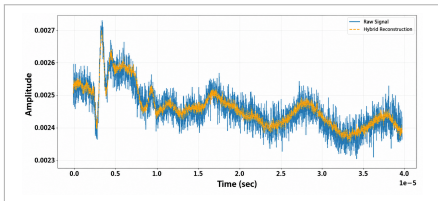
Signal Reconstruction from Hybrid Feature Set



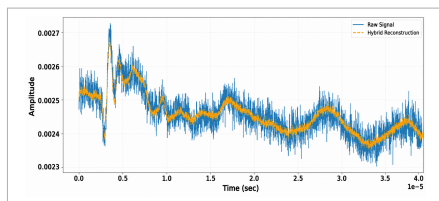
Latent = 16 + Engineered = 16



Latent = 24 + Engineered = 8



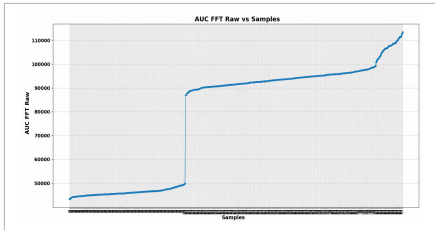
Latent = 20 + Engineered = 12



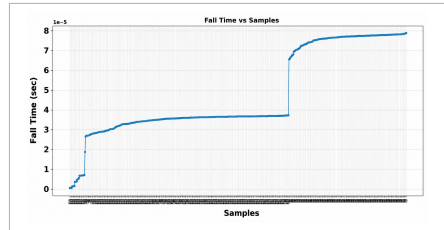
Latent = 28 + Engineered = 4

Comparison of reconstructed signals from hybrid feature set for different combinations of latent and engineered feature sets

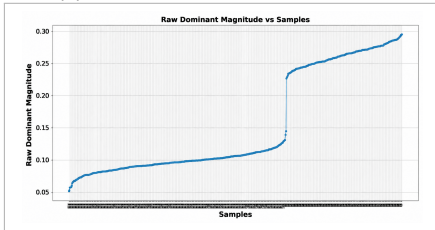
Engineered Features vs Samples



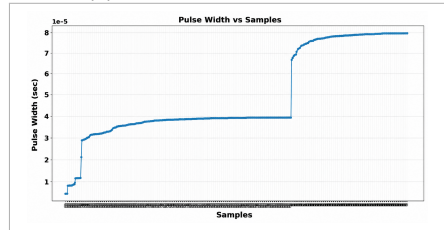
(a) AUC FFT Raw vs Samples



(b) Fall Time vs Samples



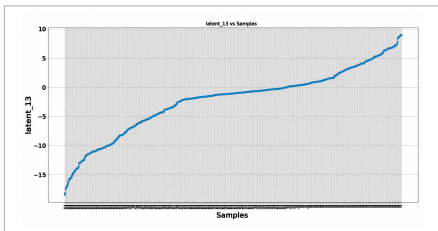
(c) Magnitude of Dominant Frequency
Component vs Samples



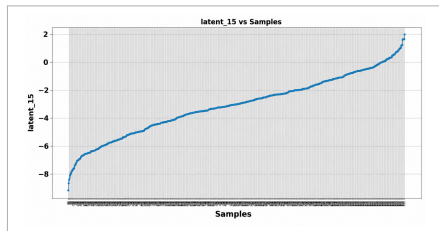
(d) Pulse Width vs Samples

Distribution of engineered features across all experimental PA signal samples

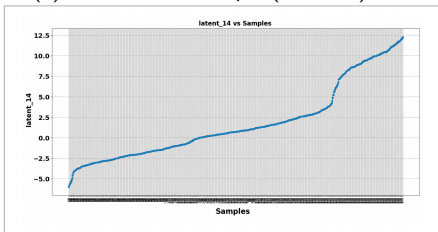
Latent Features vs Samples



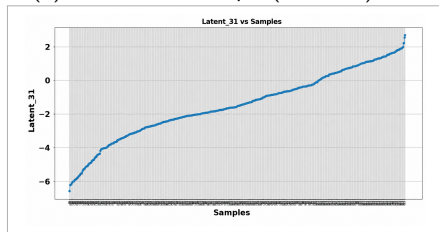
(a) Latent 13 vs Samples(Included)



(b) Latent 15 vs Samples(Excluded)



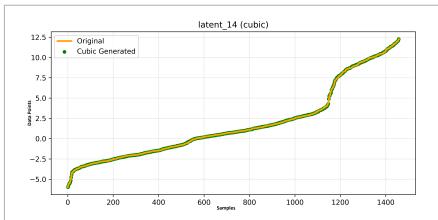
(c) Latent 14 vs Samples(Included)



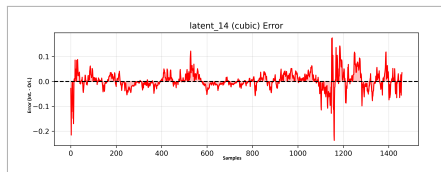
(d) Latent 31 vs Samples(Excluded)

Distribution of included and excluded latent features across all experimental PA signal samples

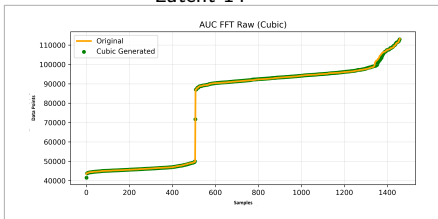
Data Densification by Interpolation



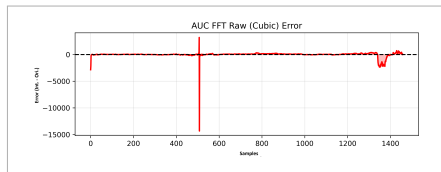
(a) Original vs Interpolated Dataset -
Latent 14



(b) Interpolation Loss - Latent 14



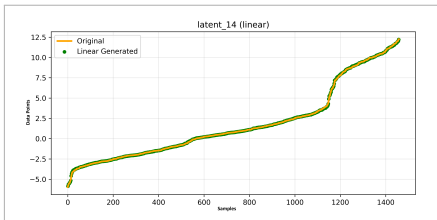
(c) Original vs Interpolated Dataset -
AUC FFT Raw



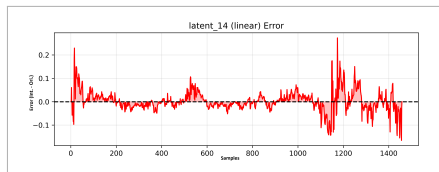
(d) Interpolation Loss - AUC FFT Raw

"CubicSpline" Interpolation Technique

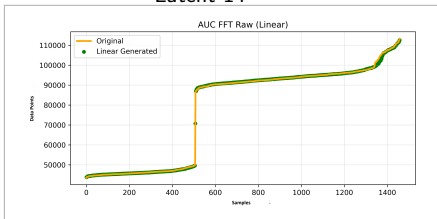
Data Densification by Interpolation



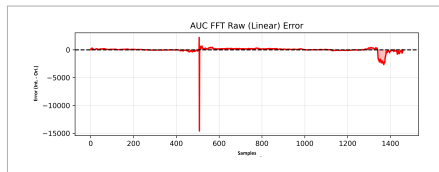
(a) Original vs Interpolated Dataset -
Latent 14



(b) Interpolation Loss - Latent 14



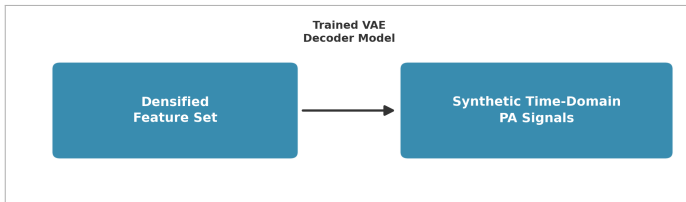
(c) Original vs Interpolated Dataset -
AUC FFT Raw



(d) Interpolation Loss - AUC FFT Raw

Linear Interpolation Technique

Synthetic Data Generation

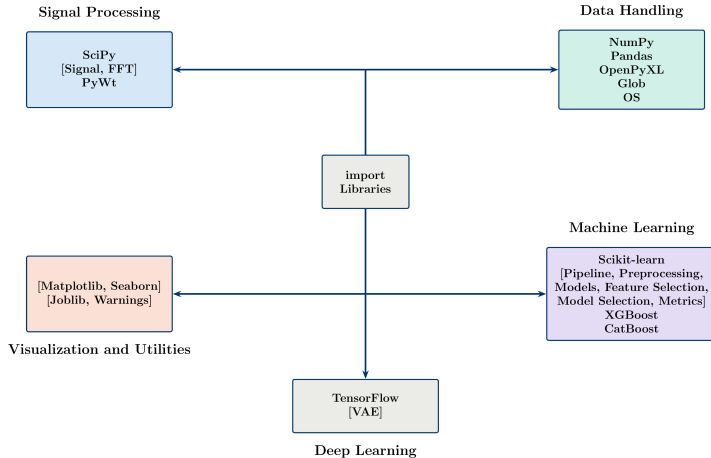


Sl. No.	Model	Training (Original Data) (Mean CV)	Testing (Original Data) Accuracy	Training (Synthetic Data) (Mean CV)	Testing (Synthetic Data) Accuracy
1	Ash Content	0.8940	90.32%	0.9189	94.26%
2	Carbon Content	0.8635	88.71%	0.8748	90.11%
3	Ignition Temperature	0.8576	87.98%	0.8746	89.21%

Future Scope

1. Extension to Assessment of Additional Coal Quality Parameters (Moisture Content, Volatile Matter, Susceptibility towards Spontaneous Combustion)
 - Fine-tune the proposed hybrid VAE-based framework for these additional coal quality parameters.
2. Multimodal Sensing Integration
 - Integrate complementary sensing modalities with PA sensing.
 - Fine-tune the proposed hybrid VAE-based framework for these additional coal quality parameters.

Python Packages Used



Key Learnings

**Python
Programming**

**Photoacoustic
Sensing**

**Spontaneous Combustion
of Coal**

**Signal
Processing**

**Feature
Engineering**

**Data
Analysis**

Regression

**Machine
Learning**

**Model
Optimization**

**Model
Evaluation**

**Scientific
Libraries**

**Research &
Documentation**

Thank You